YASHMIT SINGH

Aspiring Machine Learning Engineer | Data Science | XAI

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SUMMARY

Computer Science undergraduate specializing in Data Science and Decision Intelligence. Experienced in building explainable ML frameworks and real-time computer vision pipelines with a focus on cost efficiency and measurable impact. Actively developing a growing interest in Transformer architectures and autonomous AI systems.

TECHNICAL SKILLS

AI/ML Frameworks:	YOLOv8, XGBoost, Random Forest, Scikit-learn, SHAP (Explainability), Isotonic Regression
Data & Deployment:	FastAPI (Sub-ms Inference), Streamlit, Pandas, NumPy, Matplotlib, Seaborn, OpenCV, SQL, RESTful APIs
Programming & Web:	Python, R, C++, Java, HTML, CSS, JavaScript, Dart (Flutter)
Core Concepts:	Exploratory Data Analysis (EDA), Cost-Sensitive Learning, Data Pipelines, Feature Engineering, Model Explainability
Environment:	Git, Docker, Linux, VS Code, Ollama, OpenClaw, Local LLM Deployment

PROFESSIONAL EXPERIENCE

Weapon Detection System — **Jan 2026 – Apr 2026**

Computer Vision & Deep Learning Infrastructure

- Engineered a custom YOLOv8 model for real-time weapon detection, achieving **91.5% mAP@0.5** and **87.3% precision**.
- Optimized model training on an RTX 4060 GPU, resulting in an **88.0% recall rate** while minimizing false positives.
- Deployed a robust inference pipeline to enable low-latency, continuous threat detection across varied environmental conditions.
- Implemented dynamic confidence thresholds to ensure reliable detection for high-risk security scenarios.

Fraud Detection Framework — **Jan 2026 – Mar 2026**

Decision Intelligence & Explainable AI (XAI)

- Architected an XGBoost-based fraud detection framework, achieving a final **PR-AUC of 0.833**.
- Implemented cost-optimization strategies that reduced test costs by **11.06% ($295.75)** compared to standard baselines.
- Enhanced classification precision to **90.9%** and reduced false positives by **38.9%** (from 18 instances to 11).
- Developed a high-performance **FastAPI inference engine** with sub-millisecond latency and integrated SHAP for risk-based explainability.
- Engineered a 3-tier fallback system (Llama-3 → Mixtral → Rule-based) for automated transaction justifications.

Intern Group Leader — **Jun 2025 – Jul 2025**

Stranger Friends Helping Hands Society | NGO Leadership

- Led a 10-member team in executing community welfare campaigns, impacting underprivileged students through project-based education.
- Managed resource allocation and coordinated mentorship programs to accelerate student learning progress through structured leadership.

EDUCATION

B.Tech (Pursuing) in Computer Science (Data Science) **2024 – 2028**
University of Petroleum and Energy Studies (UPES), Dehradun
CGPA: 7.74 (Till 3rd Sem)

XII (Senior Secondary), CBSE **2024**
Rajiv International School, Mathura
Percentage: 71.2%

X (Secondary), CBSE **2022**
Rajiv International School, Mathura
Percentage: 89.9%

ACHIEVEMENTS & INTERESTS

Achievements: Student Representative for 90+ peers | Expert in Local LLM Deployment & Model Optimization (Ollama/Docker)
Interests: Transformer Architectures | Autonomous AI Agents | Financial Risk Modeling | High-Performance Computing